Revision-Note Review Worksheet for Expert Rater

This booklet contains one student revision note to review. It is a take-home revision note prepared by a teacher for a JSS1 Mathematics class on Three dimensional figures: properties of cylinders and spheres, and volume. Judge how well it covers the intended objectives and curriculum, follows the teacher's lesson plan, stays on topic and is conceptually correct, and is clear and useful for the students.

Lesson and revision-note context

Subject	Mathematics
Grade	JSS1
Term	Third Term
Week	5
Topic	Three dimensional shapes II
Subtopic	Three dimensional figures: properties of cylinders and spheres, and volume
Theme	—
Curriculum objectives	• identify properties of cylinders and spheres • find the volume of a cube, a cuboid and a prism
Previous knowledge	• properties of cubes and cuboid • properties of pyramids and cones
Lesson sequence	• identify properties of cylinders and spheres — the teacher displays shapes of cylinders and spheres; the student identify the properties; the teacher draws the shapes on the board and adds to the properties already mentioned by the students. Set for the students: 1. How many circular faces does a solid cylinder have? 2. A cylinder has how many faces. • find the volume of a cube, a cuboid and a prism — the teacher explains how to derive the formula for volume of cube and cuboid, and works the volume of a prism as area of base times height; the student listen and ask questions where they do not understand; the student copy worked examples into their notes
Instructional materials	• shapes of cylinders and spheres.
Evaluation	• Calculate the volume of a cuboid 8cm long, 4cm wide and 3cm high. • Calculate the volume of cuboid with dimensions 5m by 2m by 3m. • What is the volume of a cube of 4cm × 4cm × 4cm. • A rectangular tank hold 6m long by 2m wide holds $36m^3$ of water. How deep is the water in the tank.

Lesson objectives

1.	identify properties of cylinders and spheres
2.	find the volume of a cube, a cuboid and a prism

Read the student revision note on the following pages, then complete the rating sheet at the end of this booklet.

STUDENT REVISION NOTE

Topic: Three dimensional figures: properties of cylinders and spheres, and volume

Class: JSS1 **Subject:** Mathematics **Week:** 5

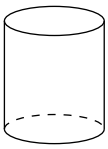
Section 1: Cylinders and Spheres: Shapes That Are Curved

Two three-dimensional shapes you meet all the time are the cylinder and the sphere. A drinks tin is a cylinder; a ball is a sphere. Even though they look quite different from the cubes and cuboids you already know, you can describe each one by its faces, edges and vertices.

Look at the cylinder in fig-01. A cylinder stands on a circular base and has a circular top. That means it has two circular faces — one on the top and one on the bottom. Around the outside, between those two circles, runs a single smooth surface that curves all the way round: the curved face. So a solid cylinder has three faces altogether — two circular and one curved.

Now count edges and vertices. The curved face meets the top circle along one edge and the bottom circle along another, so a cylinder has two curved edges. But where do the faces come together to make a point? They never do — the curved face simply wraps around. So a cylinder has no vertex.

Visual

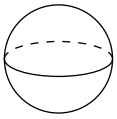


A cylinder drawn upright: an ellipse for the top face, two vertical sides falling from its ends, and a curve closing the base.

Now turn to the sphere in fig-02. A sphere is the shape of a ball — every point on its surface is the same distance from its centre. Because of that, a sphere looks identical from every direction: it is symmetrical from all directions.

Unlike the cylinder, a sphere has no flat face at all. Its whole surface is one single curved surface. And because there is nothing flat and no line where faces meet, a sphere has no edge and no vertex.

Visual



A sphere drawn as a circle with an ellipse across its middle to show the curved surface.

Practice

1. How many circular faces does a solid cylinder have?
2. How many faces does a cylinder have altogether?
3. Does a sphere have any edges or vertices?

Section 2: Volume of a Cube, a Cuboid and a Prism

Volume is the amount of space a three-dimensional shape fills, and we measure it in cubic units such as cm^3 or m^3. For a cuboid, think of the base as being made of layers that are stacked up to the height, so the volume is found by multiplying its three dimensions together:

Volume of cuboid = length × breadth × height = l × b × h

A cube is a special cuboid whose three sides are all the same length, so we simply multiply the side by itself three times:

Volume of cube = length × length × length = l^3

Calculate the volume of a cuboid of length 3cm, breadth 2cm and height 8cm.

Volume of cuboid = l × b × h
= 3cm × 2cm × 8cm
= $48cm^3$

Sometimes you know the volume and the base, and you need to find the height. The volume of a cuboid is $24cm^3$ and its base is 6cm by 2cm. Find its height.

Volume = l × b × h
$24cm^3$ = 6cm × 2cm × h
$24cm^3$ = $12cm^2$ × h

Divide both sides by $12cm^2$:
$24cm^3 \div 12cm^2$ = h
2cm = h

So the height of the cuboid is 2cm.

Calculate the volume of a cube of side 3cm.

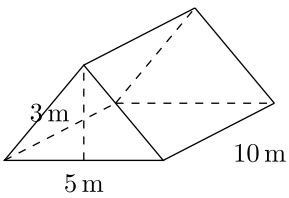
Volume of cube = l^3
= 3cm × 3cm × 3cm
= $27cm^3$

A prism has a uniform cross-section, so its volume is found by multiplying the area of its base by the height (the distance it runs back). Look at the triangular prism in fig-03. Its end is a triangle with base 5m and height 3m, and the prism runs back 10m.

First find the area of the triangular face:
Area of triangle = ½ × b × h = ½ × 5m × 3m = $7.5m^2$

Then find the volume:
Volume of prism = area of base × height
= $7.5m^2$ × 10m
= $75m^3$

Visual



A triangular prism lying on its rectangular face. The triangular end has a base of 5 m and a height of 3 m, and the prism runs back 10 m.

Practice

1. Calculate the volume of a cuboid that is 8cm long, 4cm wide and 3cm high.
2. Calculate the volume of a cuboid with dimensions 5m by 2m by 3m.
3. What is the volume of a cube of side 4cm?
4. A rectangular tank that is 6m long and 2m wide holds $36m^3$ of water. How deep is the water in the tank?

Check your work

Attempt every question before you read this page.

Cylinders and Spheres: Shapes That Are Curved

1. A solid cylinder has 2 circular faces — one on the top and one on the bottom.
2. A cylinder has 3 faces altogether: 2 circular faces and 1 curved face.
3. No. A sphere has no edge and no vertex — its whole surface is one curved surface.

Volume of a Cube, a Cuboid and a Prism

1. $96cm^3$

   Volume = l × b × h = 8cm × 4cm × 3cm = $96cm^3$.
2. $30m^3$

   Volume = l × b × h = 5m × 2m × 3m = $30m^3$.
3. $64cm^3$

   Volume = l^3 = 4cm × 4cm × 4cm = $64cm^3$.
4. 3m deep

   Volume = l × b × h, so $36m^3$ = 6m × 2m × h = $12m^2$ × h. Divide both sides by $12m^2$: h = $36m^3$ ÷ $12m^2$ = 3m.

Rating sheet

Mark one score per criterion, then use the notes section for strengths, concerns, and recommendations. The lesson objectives are on the first page for reference.

Revision note ID	cand-2e1dddcc	**Date**	
Subtopic	Three dimensional figures: properties of cylinders and spheres, and volume	**Rater**	

> The reference sheet includes the teacher's evaluation questions. A pack that reuses them, reworded or verbatim, may be credited for matching how the lesson was assessed — but reuse on its own does not establish taught-lesson alignment. Rate alignment on the whole of it: the objectives covered, the order taught, what the activities had the students do, the method, and the expected depth.

Criterion (select one score per row)	**1** Poor	**2** Weak	**3** Adequate	**4** Good	**5** Excellent
Objective and curriculum alignment Does the revision note cover the intended lesson objectives and curriculum target?	○	○	○	○	○
Taught-lesson alignment Does it reflect the taught lesson — its sequence, emphasis, activities, and expected depth?	○	○	○	○	○
Scope and conceptual correctness Does it stay within the expected lesson boundary and avoid conceptual/mathematical errors?	○	○	○	○	○
Student-facing clarity and usefulness Can JSS1 learners understand and use it meaningfully as take-home review material?	○	○	○	○	○

Strengths	Concerns	Recommendation

Flag any problems (optional)

Tick all that apply. These are separate from the scores above — leave blank if none apply.

☐	**Misses a required objective** — Skips something the lesson was meant to teach.
☐	**Goes beyond this lesson** — Includes material outside this lesson's scope (e.g. next week's topic).
☐	**Wrong or misleading content** — Contains a mathematical or factual mistake.
☐	**Too hard for JSS1 to follow** — Likely confusing for the students.
☐	**Other:** _____